



# **Partial Indistinguishability and Sector Structure in Multiphoton Interference: From the Effective Single-Particle Matrix to the Semiclassical Limit**

PhD Candidacy Examination

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Bogotá, Colombia  
2026

# Introduction

Many-photon interference in linear multiport interferometers presents two main sources of complexity. On the one hand, detection probabilities depend coherently on an increasing number of contributions as the number of photons increases. On the other hand, indistinguishability may be partial due to internal degrees of freedom that are not resolved by the measurement and that modify the way in which these contributions interfere [1, 2]. In the many-photon regime, the problem consists not only in computing individual probabilities, but also in identifying what collective structure emerges at the macroscopic scale and what quantum information is not contained in that description.

A previous result showed that, when the number of modes remains fixed and  $N \rightarrow \infty$ , part of this structure can be organized through an effective single-particle matrix and an associated classical geometry [3]. This description makes it possible to characterize macroscopic regions of the output distribution and to separate their global behavior from finer interference structures. The fact that a multiphotonic problem admits an organization determined by an object defined in the single-particle space raises, however, the need to establish what information lies outside this reduction.

Representation theory makes it possible to formulate this problem in terms of irreducible sectors and multiplicity spaces. The many-photon space can be decomposed into irreducible representations and, within them, into spaces associated with the different occupations [4, 5]. Moreover, the spectrum of the effective single-particle matrix is asymptotically related to the representations that concentrate the statistical weight as  $N \rightarrow \infty$  [6]. This relation connects the macroscopic description with the symmetry structure of the problem, but does not by itself determine the information contained within the irreducible sectors when spaces with multiplicity appear.

This proposal studies precisely the relation between these two levels of description. The focus is on determining which part of the semiclassical structure of the interference remains fixed by the effective single-particle matrix and what additional information appears when the irreducible sectors and their multiplicity spaces are resolved. The results developed so far make it possible to establish a connection between the two descriptions and show that multiplicity can retain information that is not contained solely in the single-particle description. At the same time, the incorporation of this structure into the complete sectorial probabilities and the extension of the asymptotic analysis beyond regular stationary points remain open.

**Supplementary material and computational reproducibility.** The detailed mathematical derivations supporting this proposal are provided in the companion document *Mathematical Appendices (A–I)*. The accompanying Wolfram Language code, `Multiphoton_Interference_Calculations.wl`, contains the symbolic and numerical implementations underlying the results and figures presented in both documents. These materials, together with the information required to reproduce the numerical results, are publicly available in the GitHub repository.

# 1 Physical framework and formulation of the problem

## 1.1 Multiphotonic interference with partial indistinguishability

Multiphotonic interference depends on the degree to which the photons are indistinguishable in the degrees of freedom that are not resolved by the measurement. The Hong–Ou–Mandel effect illustrates this dependence: two indistinguishable photons arriving simultaneously at a balanced beam splitter leave through the same port, so that the probability of detecting one photon at each output vanishes [7]. In multiport networks with a larger number of photons, the structure of the detection probabilities becomes considerably more complex. For perfectly indistinguishable photons, the transition amplitudes between occupations are expressed in terms of permanents of matrices constructed from the single-particle transformation [8]. When indistinguishability is partial, the overlaps between the internal states modify the interference among the different contributions to the detection probability [1, 2].

Consider an  $M$ -port interferometer, described at the single-particle level by a unitary matrix  $U \in U(M)$ . Let  $\mathbf{n} = (n_1, \dots, n_M)$  be the input occupation, with  $N = \sum_i n_i$  photons in total. We assume that the  $n_i$  photons associated with port  $i$  share a normalized internal state  $|\chi_i\rangle$ , while the detectors resolve only the number of photons in each output port. The indistinguishability is then characterized by the Gram matrix

$$\Gamma_{ij} = \langle \chi_j | \chi_i \rangle, \quad (1.1)$$

which satisfies  $\Gamma \succeq 0$  and  $\Gamma_{ii} = 1$  [1, 2].

If all internal states coincide,  $\Gamma_{ij} = 1$  for all  $i, j$  and the photons are fully indistinguishable. In the opposite limit, if the internal states associated with different ports are orthogonal,  $\Gamma = I$ . Between these two extremes, the off-diagonal elements of  $\Gamma$  contain the complex overlaps that determine the extent to which the different contributions can interfere coherently.

The probability of detecting an output occupation  $\mathbf{n}' = (n'_1, \dots, n'_M)$  will be denoted by

$$P(\mathbf{n}' | U, \mathbf{n}, \Gamma), \quad (1.2)$$

with  $\sum_i n'_i = N$ . Although exact expressions for this probability exist, the simultaneous presence of spatial and internal degrees of freedom makes it difficult to identify its structure directly as the number of photons increases [1, 2].

We are interested in the regime in which  $M$  remains fixed while  $N$  increases. We introduce the normalized occupations

$$\bar{\mathbf{n}} = \frac{\mathbf{n}}{N}, \quad \bar{\mathbf{n}}' = \frac{\mathbf{n}'}{N}, \quad (1.3)$$

and consider the limit  $N \rightarrow \infty$  while keeping these fractions fixed. In this regime, the output occupations admit a continuous description on the macroscopic scale, on which variations associated with quantum interference remain superimposed.

## 1.2 The effective single-particle matrix and the classical map

For a two-port interferometer, the asymptotic correspondence between partially distinguishable multiphotonic interference and a classical distribution can be described through a decomposition into channels associated with representations of  $SU(2)$  [9]. For multiple modes, a semiclassical analysis developed subsequently showed that this structure can be organized through a matrix defined in the single-particle space [3].

For a normalized occupation  $\bar{\mathbf{n}}$ , we define the effective single-particle matrix

$$\rho_{\bar{\mathbf{n}}, \Gamma} = [\bar{\mathbf{n}}]^{1/2} \Gamma [\bar{\mathbf{n}}]^{1/2}, \quad (1.4)$$

where  $[\bar{\mathbf{n}}]$  denotes the diagonal matrix with entries  $\bar{n}_i$ . Since  $\Gamma_{ii} = 1$ , its diagonal reproduces the initial occupation,  $\text{diag } \rho_{\bar{\mathbf{n}}, \Gamma} = \bar{\mathbf{n}}$ , and  $\text{Tr } \rho_{\bar{\mathbf{n}}, \Gamma} = 1$ . The off-diagonal elements contain the effective coherences associated with partial indistinguishability.

The matrix  $\rho_{\bar{n},\Gamma}$  does not represent the complete state of the  $N$  photons. Its importance lies in the fact that, despite being defined in the single-particle space, it determines the geometry that organizes the multiphotonic distribution in the many-photon regime.

To describe this geometry, we introduce the phase transformations  $t_\phi = \text{diag}(e^{i\phi_1}, \dots, e^{i\phi_M})$ . A global phase is irrelevant, so the independent phases form a torus  $T^{M-1}$ . After applying these phases and propagating the effective matrix through the interferometer  $U$ , we obtain

$$\mathcal{C}_{\rho,U}(\phi) = \text{diag}[U t_\phi \rho_{\bar{n},\Gamma} t_\phi^\dagger U^\dagger], \quad \mathcal{C}_{\rho,U} : T^{M-1} \rightarrow \Delta_{M-1}, \quad (1.5)$$

where  $\Delta_{M-1}$  is the simplex of normalized occupations. Its image  $\mathcal{R}_{\rho,U} = \mathcal{C}_{\rho,U}(T^{M-1})$  defines the region accessible through this map.

The classical interpretation is direct. One may consider a family of input configurations with the same relative intensities and different relative phases. Each choice of  $\phi$  produces, after the transformation  $U$ , an output intensity distribution and, therefore, a point in  $\mathcal{R}_{\rho,U}$ .

The main result of the previous analysis is that this geometry organizes the macroscopic structure of the multiphotonic distribution [3]. As  $N$  increases, occupations lying outside  $\mathcal{R}_{\rho,U}$  have exponentially small probabilities. Within this region, the relative phases that parametrize  $\mathcal{C}_{\rho,U}$  determine the smooth envelope of the distribution. The geometry of the map also determines the appearance of caustics and, on top of this structure, quantum oscillations may persist; for full-rank matrices  $\Gamma$ , however, these oscillations become negligible in the asymptotic limit.

The effective single-particle matrix thus determines part of the macroscopic structure of the interference. However, this description does not make explicit the symmetry structure of the many-photon space or the information that may remain within its different sectors. The relation between these descriptions is summarized in Fig. 1.

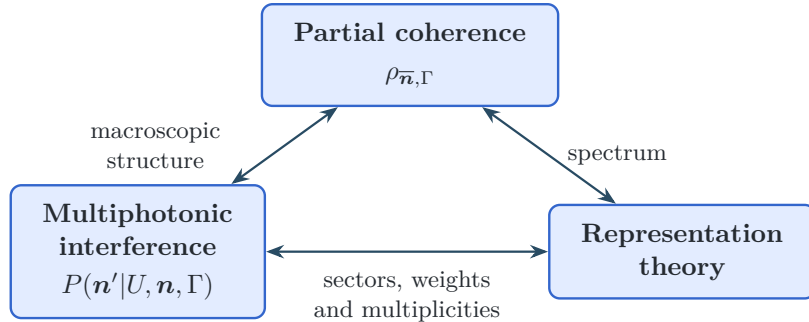


Figure 1: Relation between partial coherence, multiphotonic interference, and representation theory. The effective single-particle matrix  $\rho_{\bar{n},\Gamma}$  determines the macroscopic geometry of the multiphotonic distribution, while its spectrum makes it possible to identify asymptotically the relevant irreducible sectors and relate them to the structure of weights and multiplicities.

The connection with representation theory appears through the spectrum  $\boldsymbol{\lambda} = \text{spec } \rho_{\bar{n},\Gamma}$ . The same matrix that determines the classical map thus provides the spectral information that makes it possible to identify the relevant irreducible sectors as  $N \rightarrow \infty$ .

### 1.3 Irreducible sectors and multiplicity spaces

Schur–Weyl duality reorganizes the  $N$ -particle space into irreducible sectors,

$$(\mathbb{C}^M)^{\otimes N} = \bigoplus_{\substack{\Lambda \vdash N \\ \ell(\Lambda) \leq M}} \mathcal{H}_\Lambda \otimes \mathcal{M}_\Lambda, \quad (1.6)$$

where  $\mathcal{H}_\Lambda$  corresponds to an irreducible representation of  $U(M)$  labeled by the partition  $\Lambda = (\Lambda_1, \dots, \Lambda_M)$ , while  $\mathcal{M}_\Lambda$  contains the multiplicity associated with the action of the symmetric group



[4, 5].

The relation with the effective single-particle matrix is obtained by comparing its spectrum  $\boldsymbol{\lambda} = \text{spec } \rho_{\bar{\mathbf{n}}, \Gamma}$  with the normalized shape of a representation,

$$\bar{\Lambda} = \frac{\Lambda}{N}. \quad (1.7)$$

The result of Keyl and Werner states that, as  $N \rightarrow \infty$ , the statistical weight concentrates on representations whose normalized shape  $\bar{\Lambda}$  approaches  $\boldsymbol{\lambda}$  [6]. The relation  $\bar{\Lambda} \simeq \boldsymbol{\lambda}$  thus characterizes the sectors that dominate asymptotically, but does not imply that the two objects are equal for a finite number of photons.

Fixing a representation still does not completely determine the state relevant to the interference. Each irreducible space decomposes into weight spaces,

$$\mathcal{H}_\Lambda = \bigoplus_{\mathbf{n}} \mathcal{H}_{\Lambda, \mathbf{n}}, \quad m_\Lambda(\mathbf{n}) = \dim \mathcal{H}_{\Lambda, \mathbf{n}}, \quad (1.8)$$

where the weight  $\mathbf{n}$  fixes the occupation of the ports. When  $m_\Lambda(\mathbf{n}) > 1$ , the same occupation corresponds to several linearly independent directions within the representation.

This multiplicity is not resolved by a measurement that only counts photons in the ports. The interferometer nevertheless acts on the full  $\mathcal{H}_\Lambda$ , so the internal structure of  $\mathcal{H}_{\Lambda, \mathbf{n}}$  may modify the transition probabilities to other weights. Partial indistinguishability must therefore specify not only which sectors contribute, but also how the state is distributed within each weight space.

Once  $\Lambda$  and  $\mathbf{n}$  are fixed, this information is contained in the density

$$\sigma_{\Lambda, \mathbf{n}}(\Gamma) = \frac{P_{\Lambda, \mathbf{n}} D^\Lambda(\Gamma) P_{\Lambda, \mathbf{n}}}{\text{Tr}[P_{\Lambda, \mathbf{n}} D^\Lambda(\Gamma) P_{\Lambda, \mathbf{n}}]}, \quad (1.9)$$

defined on  $\mathcal{H}_{\Lambda, \mathbf{n}}$ . In general,  $\sigma_{\Lambda, \mathbf{n}}(\Gamma)$  is mixed and does not select a unique direction within the multiplicity space.

The detection probability can be written exactly as a sum over sectors,

$$P(\mathbf{n}' | \mathbf{n}, U, \Gamma) = \sum_{\Lambda} P(\Lambda | \mathbf{n}, \Gamma) P(\mathbf{n}' | \mathbf{n}, \Lambda, U, \Gamma), \quad (1.10)$$

$$P(\mathbf{n}' | \mathbf{n}, \Lambda, U, \Gamma) = \text{Tr} \left[ P_{\Lambda, \mathbf{n}'} D^\Lambda(U) \sigma_{\Lambda, \mathbf{n}}(\Gamma) D^\Lambda(U)^\dagger \right]. \quad (1.11)$$

This decomposition separates two aspects of the problem: the distribution of the statistical weight among representations and the dynamics within each sector. The exact derivation of this decomposition, including the Schur–Weyl reduction and the construction of  $\sigma_{\Lambda, \mathbf{n}}(\Gamma)$ , is presented in Appendix A.

Spectral concentration makes it possible to identify the relevant representations when  $N \rightarrow \infty$ , but does not determine how the dynamics is organized within a space  $\mathcal{H}_{\Lambda, \mathbf{n}}$  with multiplicity. In particular, knowing  $\boldsymbol{\lambda}$  and  $\mathbf{n}$  is not sufficient to determine the structure of  $\sigma_{\Lambda, \mathbf{n}}(\Gamma)$  or its effect on the transition probabilities. Multiplicity thus introduces an additional level of information that is not determined solely by the effective single-particle matrix.

## 2 From the spectrum to sectorial geometry: frames and projected states

### 2.1 Spectrum–diagonal compatibility and frames

Let  $\Lambda = (\Lambda_1, \dots, \Lambda_M)$  be a partition of  $N$  labeling an irreducible representation of  $U(M)$ , and define its normalized shape

$$\bar{\Lambda} = \frac{\Lambda}{N}. \quad (2.1)$$

As established in Chapter 1,  $\bar{\Lambda}$  must be distinguished from the physical spectrum  $\boldsymbol{\lambda} = \text{spec } \rho_{\bar{\mathbf{n}}, \Gamma}$ . The former characterizes a particular sector, whereas the latter is determined by the initial occupation and the indistinguishability matrix. The two approach one another for the sectors that dominate as  $N \rightarrow \infty$ , but they are not, in general, the same object.

Consider now a unitary matrix  $V \in U(M)$  and the matrix

$$\rho_{\Lambda}(V) = V[\bar{\Lambda}]V^{\dagger}, \quad [\bar{\Lambda}] = \text{diag}(\bar{\Lambda}_1, \dots, \bar{\Lambda}_M). \quad (2.2)$$

This matrix has spectrum  $\bar{\Lambda}$  and its diagonal satisfies

$$\text{diag } \rho_{\Lambda}(V) = Q(V)\bar{\Lambda}, \quad Q(V)_{ij} = |V_{ij}|^2. \quad (2.3)$$

The matrix  $Q(V)$  is doubly stochastic and, since its elements are given by  $|V_{ij}|^2$  with  $V$  unitary, it also belongs to the set of unistochastic matrices [10, 11].

For the diagonal of  $\rho_{\Lambda}(V)$  to coincide with the normalized initial occupation, we must impose

$$Q(V)\bar{\Lambda} = \bar{\mathbf{n}}. \quad (2.4)$$

We refer to the unitary matrices satisfying this condition as *compatible frames* and denote their set by

$$\mathcal{F}(\bar{\mathbf{n}}, \bar{\Lambda}) = \{V \in U(M) : Q(V)\bar{\Lambda} = \bar{\mathbf{n}}\}. \quad (2.5)$$

The shape of the representation thus fixes the spectrum, the preparation fixes the diagonal, and  $V$  determines the orientation of the eigenvectors with respect to the mode basis.

The existence of at least one compatible frame is determined by the Schur–Horn theorem. In particular, there exists a Hermitian matrix with spectrum  $\bar{\Lambda}$  and diagonal  $\bar{\mathbf{n}}$  if and only if

$$\bar{\mathbf{n}} \prec \bar{\Lambda}, \quad (2.6)$$

where  $\prec$  denotes the majorization relation [12, 13].

This condition establishes when solutions exist, but does not describe the frames that realize them. The additional difficulty arises from the fact that not every doubly stochastic matrix can be written as  $Q(V)$ . Moreover, even when  $Q(V)$  is fixed, the phases of  $V$  are not determined. This information will be important when we project the frame onto a weight space with multiplicity: two unitary matrices with the same  $Q(V)$  may select different directions within  $\mathcal{H}_{\Lambda, \mathbf{n}}$ .

The matrix  $\rho_{\Lambda}(V)$  should not, in general, be identified with the physical single-particle matrix  $\rho_{\bar{\mathbf{n}}, \Gamma}$ . The former is constructed from a sector  $\Lambda$ , whereas the latter is fixed by the preparation. They coincide when

$$\bar{\Lambda} = \boldsymbol{\lambda} = \text{spec } \rho_{\bar{\mathbf{n}}, \Gamma} \quad (2.7)$$

and  $V$  is chosen as a matrix of eigenvectors of  $\rho_{\bar{\mathbf{n}}, \Gamma}$ . This coincidence establishes the relation between the sectorial construction and the effective single-particle matrix.

## 2.2 Projected states and projection onto weights

The spectrum–diagonal condition is defined in the single-particle space. To study its effect within an irreducible representation, we consider the action of  $V$  through  $D^{\Lambda}(V)$  on  $\mathcal{H}_{\Lambda}$ .

Let  $|\Omega_{\Lambda}\rangle$  be a highest-weight vector of the representation  $\Lambda$ . The action of a frame  $V$  generates the state [14]

$$|\Omega_{\Lambda}(V)\rangle = D^{\Lambda}(V)|\Omega_{\Lambda}\rangle. \quad (2.8)$$

The explicit construction of this orbit through exterior powers and the columns of  $V$  is developed in Appendix C.

In general,  $|\Omega_{\Lambda}(V)\rangle$  contains components associated with different weights. To select the initial

occupation  $\mathbf{n}$ , we project onto  $\mathcal{H}_{\Lambda, \mathbf{n}}$ . Whenever the projection is nonzero, we define

$$|\mathbf{n}, V; \Lambda\rangle = \frac{P_{\Lambda, \mathbf{n}} D^\Lambda(V) |\Omega_\Lambda\rangle}{\sqrt{\langle \Omega_\Lambda | D^\Lambda(V)^\dagger P_{\Lambda, \mathbf{n}} D^\Lambda(V) | \Omega_\Lambda \rangle}} \in \mathcal{H}_{\Lambda, \mathbf{n}}. \quad (2.9)$$

In this way, each compatible frame selects a direction within the weight space. If  $m_\Lambda(\mathbf{n}) = 1$ , all nonzero projected states necessarily correspond to the same direction, up to a global phase. When  $m_\Lambda(\mathbf{n}) > 1$ , different frames may produce different states within  $\mathcal{H}_{\Lambda, \mathbf{n}}$ . In particular, the compatibility condition depends only on  $Q(V)$ , whereas the projected state also retains the phase information contained in  $V$ .

Projection onto a weight admits a particularly useful representation in terms of the relative phases of the maximal torus. Taking

$$t_\phi = \text{diag}(e^{i\phi_1}, \dots, e^{i\phi_{M-1}}, 1), \quad \phi \in T^{M-1}, \quad (2.10)$$

the projector can be written as

$$P_{\Lambda, \mathbf{n}} = \int_{T^{M-1}} \frac{d^{M-1}\phi}{(2\pi)^{M-1}} \exp\left[-i \sum_{j=1}^{M-1} n_j \phi_j\right] D^\Lambda(t_\phi). \quad (2.11)$$

This expression is exact: the integral selects the component of the orbit with weight  $\mathbf{n}$  and keeps explicit the dependence on the relative phases of the frame. This formulation will be useful for studying the norm and inner products of the projected states. The reduction to  $M - 1$  relative phases and the generating function associated with the norm of the projected state are developed in Appendix C.

### 2.3 Frame state and physical density

The states  $|\mathbf{n}, V; \Lambda\rangle$  make it possible to study particular directions within  $\mathcal{H}_{\Lambda, \mathbf{n}}$ , but they do not, in general, describe all the information associated with partial indistinguishability within this space. This information is contained in the density  $\sigma_{\Lambda, \mathbf{n}}(\Gamma)$  defined in Eq. (1.9).

The difference between the two can be seen directly from their ranks. The frame state determines the pure projector

$$|\mathbf{n}, V; \Lambda\rangle \langle \mathbf{n}, V; \Lambda|, \quad (2.12)$$

which has rank one. By contrast, if  $\Gamma > 0$  and the space  $\mathcal{H}_{\Lambda, \mathbf{n}}$  is nonempty, the density  $\sigma_{\Lambda, \mathbf{n}}(\Gamma)$  satisfies

$$\text{rank } \sigma_{\Lambda, \mathbf{n}}(\Gamma) = m_\Lambda(\mathbf{n}). \quad (2.13)$$

Therefore, when  $m_\Lambda(\mathbf{n}) > 1$ , a single frame state cannot by itself represent the density  $\sigma_{\Lambda, \mathbf{n}}(\Gamma)$ .

This difference can also be seen from the spectral decomposition of  $\rho_{\bar{\mathbf{n}}, \Gamma}$ . When considering its action within the representation  $\Lambda$  and projecting onto the initial weight, the density  $\sigma_{\Lambda, \mathbf{n}}(\Gamma)$  incorporates contributions associated with different weights and, when multiplicity is present, with different directions within them. The state  $|\mathbf{n}, V; \Lambda\rangle$ , by contrast, is obtained by projecting only the orbit of the highest-weight vector. The decomposition of  $\sigma_{\Lambda, \mathbf{n}}(\Gamma)$  into contributions arising from the different weights of the representation, as well as the proof of the rank property in Eq. (2.13), are developed in Appendix G.

The frame states thus form a family of pure states with which we can explore the multiplicity space. A collection of these states may linearly generate  $\mathcal{H}_{\Lambda, \mathbf{n}}$  without its elements being necessarily orthogonal or constituting a probabilistic decomposition of  $\sigma_{\Lambda, \mathbf{n}}(\Gamma)$ . This distinguishes two aspects of the problem: the directions that can be reached by varying the compatible frame and the way in which partial indistinguishability determines the density on the same space.

The three-mode case allows both aspects to be analyzed explicitly. For  $SU(3)$ , the spectrum–diagonal condition can be studied through the geometry of unistochastic matrices, whereas the multiplicity

of the weight spaces can be resolved in a basis that makes it possible to distinguish their different directions.

### 3 Spectrum–diagonal geometry and multiplicity in $SU(3)$

#### 3.1 Spectrum–diagonal geometry and reconstruction of the frames

The spectrum–diagonal condition defined in Chapter 2 is

$$Q(V)\bar{\Lambda} = \bar{n}, \quad Q(V)_{ij} = |V_{ij}|^2. \quad (3.1)$$

In the case  $M = 3$ , this condition can first be studied in terms of the squared moduli of  $V$ . To characterize the compatible matrices  $Q$  before reconstructing the phases of the frame, we consider the Birkhoff polytope of order three,

$$\mathcal{B}_3 = \left\{ Q \in \mathbb{R}^{3 \times 3} : Q_{ij} \geq 0, Q\mathbf{1} = \mathbf{1}, Q^T\mathbf{1} = \mathbf{1} \right\}, \quad (3.2)$$

whose elements are bistochastic matrices. Its vertices correspond to the six permutation matrices and  $\dim \mathcal{B}_3 = (3 - 1)^2 = 4$ .

The spectrum–diagonal condition selects within this polytope the set

$$\mathcal{P}(\bar{n}, \bar{\Lambda}) = \left\{ Q \in \mathcal{B}_3 : Q\bar{\Lambda} = \bar{n} \right\}. \quad (3.3)$$

To make its geometry explicit, we parametrize a matrix in  $\mathcal{B}_3$  by four independent entries,

$$Q(x, u, y, v) = \begin{pmatrix} x & u & 1 - x - u \\ y & v & 1 - y - v \\ 1 - x - y & 1 - u - v & x + u + y + v - 1 \end{pmatrix}, \quad \begin{aligned} x &= q_{11}, & u &= q_{12}, \\ y &= q_{21}, & v &= q_{22}. \end{aligned} \quad (3.4)$$

Since  $\bar{\Lambda}$  and  $\bar{n}$  are normalized, the equation  $Q\bar{\Lambda} = \bar{n}$  contains only two independent constraints. For  $\bar{\Lambda}_2 \neq \bar{\Lambda}_3$ , these constraints allow us to write

$$u(x) = \frac{\bar{n}_1 - \bar{\Lambda}_3 - (\bar{\Lambda}_1 - \bar{\Lambda}_3)x}{\bar{\Lambda}_2 - \bar{\Lambda}_3}, \quad v(y) = \frac{\bar{n}_2 - \bar{\Lambda}_3 - (\bar{\Lambda}_1 - \bar{\Lambda}_3)y}{\bar{\Lambda}_2 - \bar{\Lambda}_3}. \quad (3.5)$$

Substituting these expressions into (3.4), we obtain

$$Q(x, y) = \begin{pmatrix} x & u(x) & 1 - x - u(x) \\ y & v(y) & 1 - y - v(y) \\ 1 - x - y & 1 - u(x) - v(y) & x + y + u(x) + v(y) - 1 \end{pmatrix}. \quad (3.6)$$

The spectrum–diagonal condition thus reduces the four-dimensional geometry of  $\mathcal{B}_3$  to a two-dimensional region in the  $(x, y)$  plane. The conditions  $Q_{ij} \geq 0$  delimit a convex polygon in this plane, whose shape depends on  $\bar{\Lambda}$  and the initial occupation  $\bar{n}$ . The cases with degenerate spectrum and the reconstruction of the compatible frames are discussed in Appendix B.

The condition for the existence of a compatible solution can be expressed through the Schur–Horn theorem,

$$\bar{n} \prec \bar{\Lambda}, \quad (3.7)$$

where  $\prec$  denotes majorization [12, 13]. This relation guarantees the existence of a unitary matrix  $V$  such that  $\text{diag}(V[\bar{\Lambda}]V^\dagger) = \bar{n}$  and, therefore,  $Q(V)\bar{\Lambda} = \bar{n}$ .

However, not every bistochastic matrix belongs to the unistochastic set. Therefore, after determining the polygon  $\mathcal{P}(\bar{n}, \bar{\Lambda})$ , we must identify which of its points can be written as the squared moduli of a unitary matrix. For matrices of order three, this condition admits a direct geometric interpretation [10, 11, 15].

Using the row and column phase freedoms, we can take the first two columns of  $V$  in the form

$$v_1 = \begin{pmatrix} \sqrt{Q_{11}} \\ \sqrt{Q_{21}} \\ \sqrt{Q_{31}} \end{pmatrix}, \quad v_2 = \begin{pmatrix} \sqrt{Q_{12}} \\ \sqrt{Q_{22}} e^{i\phi_2} \\ \sqrt{Q_{32}} e^{i\phi_3} \end{pmatrix}. \quad (3.8)$$

The orthogonality condition  $v_1^\dagger v_2 = 0$  then becomes

$$L_1 + L_2 e^{i\phi_2} + L_3 e^{i\phi_3} = 0, \quad L_i = \sqrt{Q_{i1} Q_{i2}}, \quad i = 1, 2, 3. \quad (3.9)$$

This equation requires three complex numbers with moduli  $L_1$ ,  $L_2$ , and  $L_3$  to close a triangle in the complex plane. The phases therefore exist if and only if

$$L_1 \leq L_2 + L_3, \quad L_2 \leq L_1 + L_3, \quad L_3 \leq L_1 + L_2. \quad (3.10)$$

For bistochastic matrices of order three, these inequalities constitute a necessary and sufficient condition for unistochasticity.

This criterion can be written algebraically by defining

$$\xi_i = L_i^2 = Q_{i1} Q_{i2}, \quad i = 1, 2, 3, \quad (3.11)$$

and

$$H_{\text{Heron}}(Q) = 2(\xi_1 \xi_2 + \xi_2 \xi_3 + \xi_3 \xi_1) - (\xi_1^2 + \xi_2^2 + \xi_3^2). \quad (3.12)$$

When  $L_1$ ,  $L_2$ , and  $L_3$  form a triangle, Heron's formula gives  $H_{\text{Heron}}(Q) = 16A^2$ , where  $A$  is its area. The sign of this quantity allows the points of the polygon to be classified directly:

$$\begin{cases} H_{\text{Heron}}(Q) > 0, & Q \text{ lies in the interior of the unistochastic region,} \\ H_{\text{Heron}}(Q) = 0, & Q \text{ lies on its boundary,} \\ H_{\text{Heron}}(Q) < 0, & Q \text{ is not unistochastic.} \end{cases} \quad (3.13)$$

The subset of compatible matrices that admit a unitary frame is then

$$\mathcal{U}(\bar{\mathbf{n}}, \bar{\Lambda}) = \left\{ Q \in \mathcal{P}(\bar{\mathbf{n}}, \bar{\Lambda}) : H_{\text{Heron}}(Q) \geq 0 \right\}. \quad (3.14)$$

Fig. 2 illustrates this construction for the asymmetric configuration used in the numerical calculations. The section  $Q\bar{\Lambda} = \bar{\mathbf{n}}$  determines the polygon of bistochastic matrices, whereas the condition  $H_{\text{Heron}} \geq 0$  selects within it the region that admits unitary frames. The triangle associated with an interior point shows geometrically the closure condition that determines the phases.

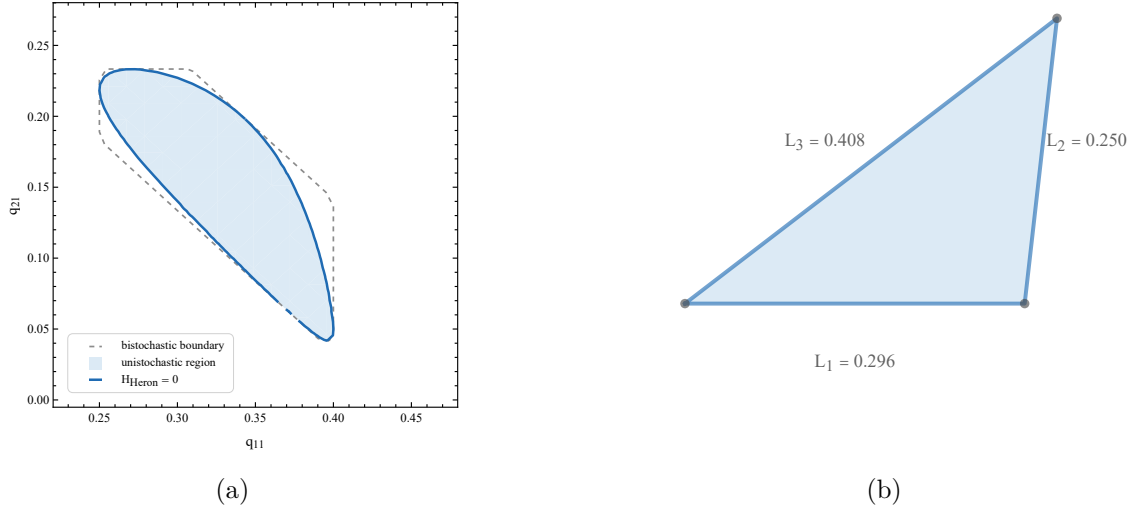


Figure 2: Bistochastic and unistochastic regions for the asymmetric configuration  $\bar{\mathbf{n}} = (0.3333, 0.2222, 0.4444)$ , with  $\bar{\Lambda} = (0.7333, 0.2000, 0.0667)$ ,  $N = 45$ , and  $\Lambda = (33, 9, 3)$ . (a) Affine section defined by  $Q\bar{\Lambda} = \bar{\mathbf{n}}$  in the  $(q_{11}, q_{21})$  plane. The dashed curve indicates the bistochastic boundary, and the light-blue region corresponds to the unistochastic subset  $H_{\text{Heron}} \geq 0$ . (b) Triangle associated with the representative frame  $V_1$ . Since  $H_{\text{Heron}} > 0$ , the corresponding point lies in the interior of the unistochastic region.

Once a point in the unistochastic region has been identified, the same construction allows the phases of the frame to be reconstructed. For an interior point with  $L_1 L_2 \neq 0$ , the law of cosines applied to (3.9) gives

$$\cos \phi_2 = \frac{L_3^2 - L_1^2 - L_2^2}{2L_1 L_2}, \quad (3.15)$$

from which the two solutions are obtained

$$\phi_2^{(\pm)} = \pm \arccos\left(\frac{L_3^2 - L_1^2 - L_2^2}{2L_1 L_2}\right), \quad \phi_3^{(\pm)} = \arg\left[-\left(L_1 + L_2 e^{i\phi_2^{(\pm)}}\right)\right]. \quad (3.16)$$

These two choices determine two unitary branches  $V_+$  and  $V_-$  which, after fixing the phases that do not affect  $Q$ , are related by complex conjugation and satisfy

$$|V_+|^{\circ 2} = |V_-|^{\circ 2} = Q. \quad (3.17)$$

On the boundary  $H_{\text{Heron}}(Q) = 0$ , the triangle degenerates and the two branches coincide.

The matrix  $Q$  thus determines the spectrum–diagonal compatibility and the moduli of the frame elements, but does not completely determine their phases. In the interior of the unistochastic region, two frames may exist with the same matrix  $Q$  and different phase information. When projected onto a weight with multiplicity, this difference may give rise to distinct states within  $\mathcal{H}_{\Lambda, n}$ .

Consider a representation of  $U(3)$  written as

$$\Lambda = (p + q + r, q + r, r), \quad p = \Lambda_1 - \Lambda_2, \quad q = \Lambda_2 - \Lambda_3, \quad r = \Lambda_3. \quad (3.18)$$

For  $SU(3)$ , the common shift determined by  $r$  can be separated and the representation is characterized by  $(p, q)$ . We then introduce the reduced weight

$$W = \mathbf{n} - r\mathbf{1}. \quad (3.19)$$

Since  $Q\mathbf{1} = \mathbf{1}$ , the compatibility condition can equivalently be written as

$$Q \begin{pmatrix} p+q \\ q \\ 0 \end{pmatrix} = W. \quad (3.20)$$

The relation between the partition of  $U(3)$ , the irrep  $(p, q)$  of  $SU(3)$ , and the explicit construction of the weights is developed in Appendix D.

The weight spaces of a representation of  $SU(3)$  can be resolved using the chain

$$U(3) \supset U(2) \supset U(1), \quad (3.21)$$

[16]. For a fixed weight  $W$ , the different directions can be labeled by an isospin number  $j$ . We denote the corresponding basis by

$$\{|W, j\rangle\}. \quad (3.22)$$

The allowed values of  $j$  lie between

$$j_{\min} = \frac{1}{2} \max\{|W_1 - W_2|, |p - W_3|\}, \quad (3.23)$$

$$j_{\max} = \frac{1}{2} \min\{p + W_3, p + 2q - W_3\}, \quad (3.24)$$

and increase in integer steps. The multiplicity of the weight is therefore

$$m_{\Lambda}(\mathbf{n}) = j_{\max} - j_{\min} + 1, \quad (3.25)$$

provided that the interval is nonempty. The derivation of these bounds from the Gelfand–Tsetlin patterns and their interlacing conditions is presented in Appendix E.

We thus obtain an orthonormal basis

$$\mathcal{H}_{\Lambda, \mathbf{n}} = \text{span} \{|W, j\rangle : j_{\min} \leq j \leq j_{\max}\}. \quad (3.26)$$

The multiplicity can then be described in terms of concrete directions labeled by  $j$ , which makes it possible to compare directly the states obtained from different frames.

### 3.2 Frame states in the isospin basis

For a frame  $V$  satisfying the spectrum–diagonal condition, the projected state defined in Eq. (2.9) belongs to  $\mathcal{H}_{\Lambda, \mathbf{n}}$ . Using the isospin basis of Eq. (3.26), it can be written as

$$|W, V\rangle \equiv |\mathbf{n}, V; \Lambda\rangle = \sum_{j=j_{\min}}^{j_{\max}} c_j(V) |W, j\rangle, \quad \sum_j |c_j(V)|^2 = 1. \quad (3.27)$$

The coefficients  $c_j(V)$  determine the coordinates of the projected state within the multiplicity space. These coordinates can be obtained exactly. If  $B_j(V)$  denotes the unnormalized coefficient associated with the isospin value  $j$ , then

$$c_j(V) = \frac{B_j(V)}{\sqrt{\sum_k |B_k(V)|^2}}. \quad (3.28)$$

The coefficients  $B_j(V)$  can be expressed in terms of Wigner- $D$  elements. To obtain this form, the polynomial representation of the matrices  $D^j$  and the formulas for the Clebsch–Gordan coefficients are used [17, 18]. The explicit derivation of the coefficients  $B_j(V)$  and  $c_j(V)$  in terms of Wigner- $D$  matrices and Clebsch–Gordan coefficients is developed in Appendix E.

The expansion (3.27) also shows what information from the frame remains after projection. The matrix  $Q(V)$  determines its squared moduli and the spectrum–diagonal condition, but does not fix the phases of  $V$ . These phases enter the coefficients  $c_j(V)$  and, therefore, may change the direction selected within  $\mathcal{H}_{\Lambda, \mathbf{n}}$  even for frames with the same matrix  $Q(V)$ .

The squared moduli of these coordinates,

$$P_{\text{orb}}(j|W, V) = |c_j(V)|^2, \quad (3.29)$$

describe the distribution of the projected state among the different isospin values. This distribution corresponds to the pure state associated with the frame and, in general, does not coincide with the diagonal elements of  $\sigma_{\Lambda, \mathbf{n}}(\Gamma)$  in the same basis. The comparison between the two distributions is developed in Appendix G.

The same expansion allows us to associate with each frame the vector

$$c(V) = (c_j(V))_{j_{\min} \leq j \leq j_{\max}} \in \mathbb{C}^{m_{\Lambda}(\mathbf{n})}. \quad (3.30)$$

In this way, the states obtained from different frames can be compared directly through the Hermitian inner products of their coordinate vectors. For a family of frames, this representation makes it possible to determine how many linearly independent directions are generated within  $\mathcal{H}_{\Lambda, \mathbf{n}}$  and whether, taken together, they generate the entire multiplicity space.

### 3.3 Inner products and generation of the multiplicity space

The coordinates  $c_j(V)$  allow the states associated with different frames to be compared directly. For two frames  $V$  and  $V'$ , we define

$$K_{\Lambda, \mathbf{n}}(V, V') = \langle \mathbf{n}, V; \Lambda | \mathbf{n}, V'; \Lambda \rangle = \sum_j c_j(V)^* c_j(V'). \quad (3.31)$$

The kernel can also be computed directly from the frames through coefficient extraction from the generating function of the projected orbits. The representation of the kernel through the generating function of the projected orbits and its normalized formulation are developed in Appendix F.

Consider now a finite family of frames  $V_1, \dots, V_K$  and collect their coordinate vectors into

$$C = (c(V_1) \ c(V_2) \ \cdots \ c(V_K)). \quad (3.32)$$

The Gram matrix of the projected states is

$$G = C^\dagger C, \quad G_{ab} = \langle \mathbf{n}, V_a; \Lambda | \mathbf{n}, V_b; \Lambda \rangle. \quad (3.33)$$

Therefore,

$$\text{rank } G = \dim \text{span} \{ |\mathbf{n}, V_a; \Lambda \rangle \}_{a=1}^K \leq m_{\Lambda}(\mathbf{n}). \quad (3.34)$$

A family generates the entire multiplicity space if and only if

$$\text{rank } G = m_{\Lambda}(\mathbf{n}). \quad (3.35)$$

The full-rank condition establishes that the family under consideration linearly generates  $\mathcal{H}_{\Lambda, \mathbf{n}}$ , but does not imply that every direction in this space can be obtained from a single frame. On the other hand, if a finite sample produces a rank smaller than  $m_{\Lambda}(\mathbf{n})$ , the result characterizes only that particular family and does not establish a restriction on the complete set of compatible frames. Finally, the rank of the Gram matrix determines the dimension of the generated subspace, but not the orthogonality of its elements; a family may reach full rank without forming an orthogonal basis.



### 3.4 Preliminary results

To explore numerically the family of projected states, we consider

$$N = 45, \quad \Lambda = (33, 9, 3), \quad \bar{\Lambda} = \left( \frac{11}{15}, \frac{1}{5}, \frac{1}{15} \right). \quad (3.36)$$

For the configurations studied, we also chose

$$\lambda = \text{spec } \rho_{\bar{n}, \Gamma} = \bar{\Lambda}, \quad (3.37)$$

an equality that pertains only to these examples and does not constitute a general condition.

We considered the occupations

| $\mathbf{n}$   | $W$            | $j$                                             | $m_{\Lambda}(\mathbf{n})$ |
|----------------|----------------|-------------------------------------------------|---------------------------|
| $(15, 15, 15)$ | $(12, 12, 12)$ | $6, 7, \dots, 12$                               | 7                         |
| $(27, 9, 9)$   | $(24, 6, 6)$   | $9, 10, \dots, 15$                              | 7                         |
| $(15, 10, 20)$ | $(12, 7, 17)$  | $\frac{7}{2}, \frac{9}{2}, \dots, \frac{19}{2}$ | 7                         |

(3.38)

The three weights have multiplicity seven, although they correspond to different sets of isospin values. For finite banks of compatible frames, the rank of the Gram matrix makes it possible to track the number of linearly independent directions generated as new states are added. In the representative banks analyzed, full rank was reached,

$$\text{rank } G_7 = m_{\Lambda}(\mathbf{n}) = 7. \quad (3.39)$$

For the representative ordering shown in Appendix F,  $\text{rank } G_K = K$  for  $K \leq 7$ . These calculations provide numerical evidence of complete generation for the configurations studied, but do not establish this property for an arbitrary representation, weight, or family of frames.

The role of the information not contained in  $Q(V)$  can be isolated through an exact example. Consider the adjoint representation

$$(p, q) = (1, 1), \quad \Lambda = (2, 1, 0), \quad (3.40)$$

whose central weight  $W = (1, 1, 1)$  has multiplicity two. Consider the Fourier frame

$$F_3 = \frac{1}{\sqrt{3}} \begin{pmatrix} 1 & 1 & 1 \\ 1 & \omega & \omega^2 \\ 1 & \omega^2 & \omega \end{pmatrix}, \quad \omega = e^{2\pi i/3}. \quad (3.41)$$

The frame obtained by interchanging its second and third columns has the same matrix of squared moduli,

$$Q(V) = \frac{1}{3} \mathbf{1}\mathbf{1}^T. \quad (3.42)$$

However, the normalized projections of the two frames onto the central weight are

$$|\psi_1\rangle = \frac{1}{\sqrt{3}}(1, \omega, \omega^2), \quad |\psi_2\rangle = \frac{1}{\sqrt{3}}(1, \omega^2, \omega), \quad \langle \psi_1 | \psi_2 \rangle = 0. \quad (3.43)$$

Two frames with the same matrix  $Q(V)$  can therefore select orthogonal directions within the same multiplicity space.

Fig. 3 shows this structure for the adjoint representation. For the bank considered, two linearly independent states suffice to reach the dimension  $m_{\Lambda}(\mathbf{n}) = 2$ , while the Fourier example makes explicit that the phase information of the frame can distinguish directions that  $Q(V)$  does not distinguish.

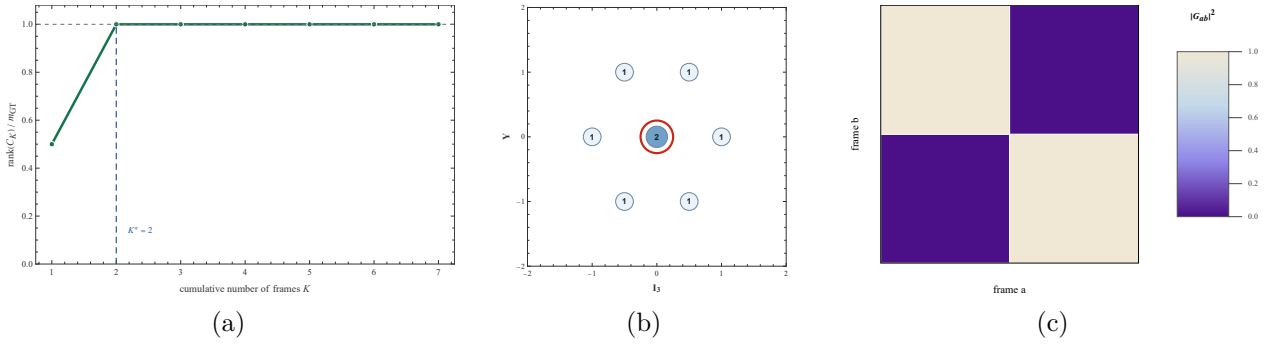


Figure 3: Multiplicity structure for the adjoint representation  $\Lambda = (2, 1, 0)$  at  $\bar{n} = (1/3, 1/3, 1/3)$ . (a) Cumulative rank for the seven-frame bank, with full rank reached at  $K_* = 2$ . (b) Weight diagram with the central weight  $W = (1, 1, 1)$  highlighted. (c) Fidelity matrix  $|G_{ab}|^2$  for two frames with the same  $Q(V)$ ; the off-diagonal zeros show that their projected states are orthogonal.

The complete generation observed for  $\Lambda = (33, 9, 3)$  and the growth of the rank for finite banks constitute numerical results for the configurations considered. By contrast, the orthogonality of the two states of the adjoint representation and the equality of their matrices  $Q(V)$  are exact. None of these results implies that every direction of  $\mathcal{H}_{\Lambda, n}$  can be obtained from a single frame.

Taken together, these examples show that  $Q(V)$  determines the spectrum–diagonal compatibility, but does not by itself specify the projected state when multiplicity is present. The information contained in the phases of the frame remains after projection and determines the direction selected within  $\mathcal{H}_{\Lambda, n}$ .

## 4 Sectorial dynamics and emergence of the classical map

### 4.1 Sectorial probability and stationarity condition

Let us fix an irrep  $\Lambda$ , an initial occupation  $\mathbf{n}$ , and a frame  $V$  satisfying the spectrum–diagonal condition. The projected pure state is

$$|\mathbf{n}, V; \Lambda\rangle = \frac{P_{\Lambda, \mathbf{n}} D^\Lambda(V) |\Omega_\Lambda\rangle}{\sqrt{\mathcal{N}_{\Lambda, \mathbf{n}}(V)}}, \quad (4.1)$$

and, after the action of an interferometer  $U$ , the probability of detecting an occupation  $\mathbf{n}'$  is given by

$$P_{\text{frame}}(\mathbf{n}' | \mathbf{n}, \Lambda, U, V) = \left\| P_{\Lambda, \mathbf{n}'} D^\Lambda(U) |\mathbf{n}, V; \Lambda\rangle \right\|^2. \quad (4.2)$$

This expression describes the dynamics of a pure frame state within a fixed irrep and, in general, does not coincide with the probability conditioned on the sector determined by  $\sigma_{\Lambda, \mathbf{n}}(\Gamma)$ . The asymptotic limits of representation functions of  $SU(2)$  and  $SU(3)$  have been studied previously in optical interference problems [19, 20].

To study its behavior in the many-photon regime, we consider sequences for which

$$\frac{\Lambda}{N} \longrightarrow \bar{\Lambda}, \quad \frac{\mathbf{n}}{N} \longrightarrow \bar{\mathbf{n}}, \quad \frac{\mathbf{n}'}{N} \longrightarrow \bar{\mathbf{n}}'. \quad (4.3)$$

The toroidal representation of the projector makes it possible to analyze the normalization of the state in this limit:

$$\mathcal{N}_{\Lambda, \mathbf{n}}(V) = \int \frac{d^M \varphi}{(2\pi)^M} e^{-i\mathbf{n} \cdot \varphi} \prod_{r=1}^M \det(V_r^\dagger t_\varphi V_r)^{p_r}, \quad p_r = \Lambda_r - \Lambda_{r+1}, \quad (4.4)$$

where  $\Lambda_{M+1} = 0$  and  $V_r$  contains the first  $r$  columns of  $V$ . The toroidal representation of  $\mathcal{N}_{\Lambda, \mathbf{n}}(V)$  follows from the projection onto weights developed in Appendix C.

For  $N \rightarrow \infty$ , with  $\bar{p}_r = p_r/N = \bar{\Lambda}_r - \bar{\Lambda}_{r+1}$ , we introduce the function

$$s(\boldsymbol{\varphi}) = -i\bar{\mathbf{n}} \cdot \boldsymbol{\varphi} + \sum_{r=1}^M \bar{p}_r \log \det \left( V_r^\dagger t_\varphi V_r \right). \quad (4.5)$$

This function contains the dominant dependence of the integral on  $N$  and makes it possible to analyze its behavior through stationary points. Evaluating the gradient of  $s$  at the origin gives

$$\nabla s(0) = i \left[ Q(V) \bar{\Lambda} - \bar{\mathbf{n}} \right], \quad \implies \quad Q(V) \bar{\Lambda} = \bar{\mathbf{n}} \iff \nabla s(0) = 0. \quad (4.6)$$

The spectrum–diagonal condition thus acquires a direct semiclassical interpretation: the same frames that relate  $\bar{\Lambda}$  to the initial occupation  $\bar{\mathbf{n}}$  determine the stationary points of the integral that normalizes the projected state.

Only  $M - 1$  phases are independent, since the global phase does not modify the projection. The stationary analysis can therefore be carried out over the  $M - 1$  relative phases. If the stationary point is regular, with reduced Hessian  $H_{\text{red}} > 0$ , the Gaussian approximation gives

$$\mathcal{N}_{\Lambda, \mathbf{n}}(V) \sim \frac{1}{(2\pi N)^{(M-1)/2} \sqrt{\det H_{\text{red}}}}, \quad -\frac{1}{N} \log \mathcal{N}_{\Lambda, \mathbf{n}}(V) \longrightarrow 0. \quad (4.7)$$

The normalization therefore has an algebraic behavior in  $N$  and does not introduce an additional exponential rate into the transition probability. The exponential dependence is determined by the projection onto the final occupation. The calculation of the gradient, the reduced Hessian, and the regular Gaussian approximation is developed in Appendix H.

## 4.2 Large deviations and the sectorial classical map

The construction of the irrep through exterior powers makes it possible to reorganize the amplitudes into rank- $k$  blocks, each repeated  $p_k = \Lambda_k - \Lambda_{k+1}$  times. In the regime  $N \rightarrow \infty$ , we define  $\alpha_k = p_k/N$ , so that  $\sum_{k=a}^M \alpha_k = \bar{\Lambda}_a$ . These coefficients determine the relative weight of each block in the asymptotic behavior.

For a choice of relative phases  $\phi$ , let  $g_\phi = U t_\phi V$ . The action of  $g_\phi$  on an exterior block of rank  $k$  can be written as

$$\bigwedge^k g_\phi |\Omega_k\rangle = \sum_{|I|=k} A_I^{(k)}(\phi) |I\rangle, \quad q_I^{(k)}(\phi) = \left| A_I^{(k)}(\phi) \right|^2, \quad (4.8)$$

where  $A_I^{(k)}(\phi)$  is the minor associated with the subset  $I$ . The unitarity of  $g_\phi$  guarantees that the quantities  $q_I^{(k)}(\phi)$  form a normalized distribution for each block.

In this regime, a macroscopic output occupation can be described in terms of frequencies  $f_I^{(k)}$  that reproduce the final occupation according to

$$\bar{n}'_i = \sum_{k=1}^M \alpha_k \sum_{\substack{|I|=k \\ i \in I}} f_I^{(k)}. \quad (4.9)$$

The  $f_I^{(k)}$  thus specify how the configurations of each block must be distributed in order to produce an occupation  $\bar{\mathbf{n}}'$ .

The multinomial expansion shows that a contribution associated with these frequencies has the asymptotic behavior

$$|C_f(\phi)| = B_N(f) \exp \left[ -\frac{N}{2} \sum_{k=1}^M \alpha_k D_{\text{KL}} \left( f^{(k)} \| q^{(k)}(\phi) \right) \right] \left[ 1 + O(N^{-1}) \right], \quad (4.10)$$

where  $B_N(f)$  collects the factors that do not modify the exponential rate. The exponent compares the frequencies required to produce  $\bar{\mathbf{n}}'$  with the distributions  $q^{(k)}(\phi)$  determined by the frame, the relative phases, and the interferometer. The appearance of the relative divergence has the characteristic form of large-deviation principles for empirical frequencies [21, 22]. The exact expansion into exterior blocks, the passage to the multinomial limit, and the derivation of this rate function are developed in Appendix I.

Since the relative divergence is nonnegative, the rate can vanish only when  $f_I^{(k)} = q_I^{(k)}(\phi) = |A_I^{(k)}(\phi)|^2$ . For these frequencies, the final occupation satisfies

$$\begin{aligned}\bar{n}'_i &= \sum_{k=1}^M \alpha_k \sum_{\substack{|I|=k \\ i \in I}} \left| A_I^{(k)}(\phi) \right|^2 \\ &= \sum_{k=1}^M \alpha_k \sum_{a=1}^k |(g_\phi)_{ia}|^2 = \sum_{a=1}^M \bar{\Lambda}_a |(g_\phi)_{ia}|^2.\end{aligned}\quad (4.11)$$

The zero-rate condition thus reduces to

$$\bar{\mathbf{n}}' = Q(U t_\phi V) \bar{\Lambda}. \quad (4.12)$$

This relation defines the sectorial classical map

$$\mathcal{C}_{\bar{\Lambda}, V, U} : T^{M-1} \longrightarrow \Delta_{M-1}, \quad \mathcal{C}_{\bar{\Lambda}, V, U}(\phi) = Q(U t_\phi V) \bar{\Lambda} = \text{diag} \left[ U t_\phi V [\bar{\Lambda}] V^\dagger t_\phi^\dagger U^\dagger \right]. \quad (4.13)$$

Its image,  $\mathcal{R}_{\bar{\Lambda}, V, U} = \mathcal{C}_{\bar{\Lambda}, V, U}(T^{M-1})$ , collects the macroscopic occupations obtained by varying the relative phases.

Since the unitary transformation preserves the spectrum of  $V[\bar{\Lambda}]V^\dagger$ , these occupations satisfy the Schur–Horn condition and, therefore,

$$\mathcal{R}_{\bar{\Lambda}, V, U} \subseteq \mathcal{S}_{\bar{\Lambda}} = \left\{ \boldsymbol{\nu} \in \Delta_{M-1} : \boldsymbol{\nu} \prec \bar{\Lambda} \right\}. \quad (4.14)$$

The region  $\mathcal{S}_{\bar{\Lambda}}$  is determined solely by the shape of the representation, whereas  $\mathcal{R}_{\bar{\Lambda}, V, U}$  also depends on the frame and the interferometer.

This geometry can be expressed directly through the rate function. Let  $\mathcal{F}(\bar{\mathbf{n}}')$  denote the set of normalized frequencies that satisfy (4.9). For a fixed phase  $\phi$ , we define

$$\begin{aligned}I_\phi(\bar{\mathbf{n}}') &= \inf_{\{f^{(k)}\} \in \mathcal{F}(\bar{\mathbf{n}}')} \sum_{k=1}^M \alpha_k D_{\text{KL}} \left( f^{(k)} \| q^{(k)}(\phi) \right), \\ I(\bar{\mathbf{n}}') &= \inf_{\phi \in T^{M-1}} I_\phi(\bar{\mathbf{n}}').\end{aligned}\quad (4.15)$$

The nonnegativity of the relative divergence then implies

$$I(\bar{\mathbf{n}}') = 0 \iff \bar{\mathbf{n}}' \in \mathcal{R}_{\bar{\Lambda}, V, U}. \quad (4.16)$$

Fig. 4 shows this geometry for twelve frames satisfying the same spectrum–diagonal condition,  $Q(V)\bar{\Lambda} = \bar{\mathbf{n}}$ . Although  $\bar{\Lambda}$ ,  $\bar{\mathbf{n}}$ , and the interferometer  $U$  remain fixed, the image  $\mathcal{R}_{\bar{\Lambda}, V, U}$  changes with the choice of frame. The spectrum–diagonal condition thus fixes the initial compatibility, but does not by itself determine the geometry of the sectorial region after the interferometer.

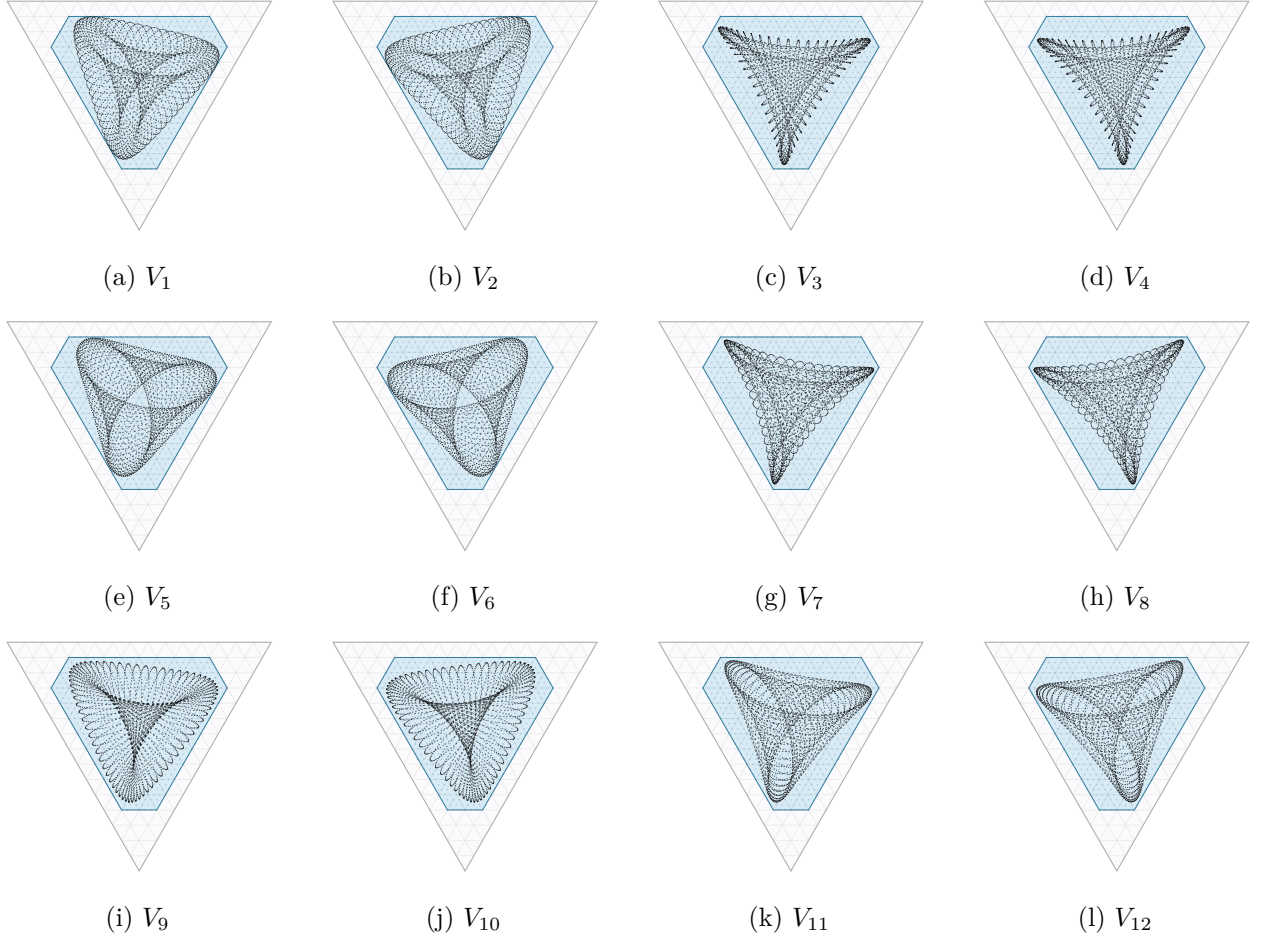


Figure 4: Images of the sectorial classical map for twelve frames satisfying  $Q(V_a)\bar{\Lambda} = \bar{\mathbf{n}}$  in the asymmetric configuration  $\bar{\mathbf{n}} = (0.3333, 0.2222, 0.4444)$ , with  $\bar{\Lambda} = (0.7333, 0.2000, 0.0667)$ ,  $\Lambda = (33, 9, 3)$ , and  $U = F_3$ . The black points represent  $\mathcal{R}_{\bar{\Lambda}, V_a, U}$ , the light-blue region corresponds to  $\mathcal{S}_{\bar{\Lambda}}$ , and the outer triangle represents the normalized occupation simplex  $\Delta_2$ .

The figure shows that the condition  $Q(V)\bar{\Lambda} = \bar{\mathbf{n}}$  does not by itself determine the geometry of the sectorial region after the interferometer. The twelve frames share the same initial occupation and the same shape  $\bar{\Lambda}$ , but generate different images  $\mathcal{R}_{\bar{\Lambda}, V, U}$  within a common spectral region.

This characterization has a direct consequence for the probabilities. If a sequence satisfies  $\mathbf{n}'_N/N \rightarrow \bar{\mathbf{n}}'$  and its limit remains at a positive distance from  $\mathcal{R}_{\bar{\Lambda}, V, U}$ , then, under the regular conditions of the analysis, there exists  $c > 0$  such that

$$P_{\text{frame}}(\mathbf{n}'_N | \mathbf{n}, \Lambda, U, V) \leq \text{poly}(N) e^{-Nc}, \quad (4.17)$$

for sufficiently large  $N$ . Occupations lying outside  $\mathcal{R}_{\bar{\Lambda}, V, U}$  therefore have exponentially small probabilities in the many-photon regime. The derivation of the map in Eq. (4.13) and the proof of exponential suppression outside  $\mathcal{R}_{\bar{\Lambda}, V, U}$  are presented in Appendix I.

For  $\bar{\mathbf{n}}' \in \mathcal{R}_{\bar{\Lambda}, V, U}$ , the exponential rate may vanish, but the map does not by itself determine the complete probability. In this region, the prefactors, the phases, and the possible contribution of several stationary points still play a role. The region  $\mathcal{R}_{\bar{\Lambda}, V, U}$  thus determines the macroscopic geometry associated with zero rate, whereas the fine structure of the distribution requires additional information.

### 4.3 Relation to the effective single-particle matrix

We introduce the matrix associated with the sector  $\rho_\Lambda(V) = V[\bar{\Lambda}]V^\dagger$ . With this notation, the map can be written as

$$\mathcal{C}_{\bar{\Lambda},V,U}(\phi) = \text{diag}\left[U t_\phi \rho_\Lambda(V) t_\phi^\dagger U^\dagger\right]. \quad (4.18)$$

This form allows it to be compared directly with the effective single-particle matrix  $\rho_{\bar{n},\Gamma} = [\bar{n}]^{1/2} \Gamma [\bar{n}]^{1/2}$  defined in Eq. (1.4).

In general, the two matrices contain different information:  $\rho_\Lambda(V)$  is associated with a particular irrep and a frame, whereas  $\rho_{\bar{n},\Gamma}$  is determined by the initial occupation and the indistinguishability matrix. They coincide exactly when the normalized shape of the irrep is equal to the spectrum of the single-particle matrix and  $V$  diagonalizes it,

$$\bar{\Lambda} = \boldsymbol{\lambda} = \text{spec } \rho_{\bar{n},\Gamma}, \quad \rho_{\bar{n},\Gamma} = V[\bar{\Lambda}]V^\dagger. \quad (4.19)$$

Under these conditions,

$$\mathcal{C}_{\bar{\Lambda},V,U}(\phi) = \text{diag}\left[U t_\phi \rho_{\bar{n},\Gamma} t_\phi^\dagger U^\dagger\right] = \mathcal{C}_{\rho,U}(\phi). \quad (4.20)$$

The sectorial map thus coincides with the map obtained for partially distinguishable multiphotonic interference [3]. If the spectrum has degeneracies, the choice of  $V$  within the degenerate subspaces does not modify this equality, since  $V[\bar{\Lambda}]V^\dagger$  continues to represent the same matrix  $\rho_{\bar{n},\Gamma}$ . The expression of the sectorial map in terms of the effective single-particle matrix is derived in Appendix I.

The equality  $\bar{\Lambda} = \boldsymbol{\lambda}$  does not hold, in general, for every sector at finite photon number. However, the Keyl–Werner result establishes that, as  $N \rightarrow \infty$ , the statistical weight concentrates on representations whose normalized shape approaches the spectrum of  $\rho_{\bar{n},\Gamma}$  [6],

$$\bar{\Lambda} \simeq \boldsymbol{\lambda} = \text{spec } \rho_{\bar{n},\Gamma}. \quad (4.21)$$

The geometry described by the sectorial map is thus related, for the sectors that dominate asymptotically, to the macroscopic geometry determined by the effective single-particle matrix.

This connection does not imply that the complete dynamics within each sector is determined by the single-particle matrix. The map was obtained for the pure state  $|\mathbf{n}, V; \Lambda\rangle$ , whereas partial indistinguishability determines, in general, a density  $\sigma_{\Lambda,\mathbf{n}}(\Gamma)$  on  $\mathcal{H}_{\Lambda,\mathbf{n}}$ . When multiplicity is present, this density cannot in general be reduced to a single frame state. To recover the complete probability, it is necessary to incorporate this structure within each sector and subsequently combine the different irreducible sectors with their respective statistical weights. The relation between the semiclassical description of the frames, the density  $\sigma_{\Lambda,\mathbf{n}}(\Gamma)$ , and the sum over sectors remains open.

The asymptotic analysis developed in Sections 4.1 and 4.2 covers regular stationary points and determines their exponential rate, but not the prefactors or the interference between different stationary contributions. When the Jacobian of  $\mathcal{C}_{\bar{\Lambda},V,U}$  loses rank, or the reduced Hessian becomes degenerate, one enters a caustic regime in which the Gaussian approximation ceases to be valid. The treatment of these regions and of possible complex stationary points requires an extension of the analysis.

Taken together, the effective single-particle matrix determines the macroscopic geometry that reappears in the dominant irreducible sectors, whereas multiplicity and the structure of the stationary points contain additional information that is not fixed solely by  $\rho_{\bar{n},\Gamma}$ .

## 5 Research design

### 5.1 Research problem

The results developed in Chapters 1–4 show that, in the many-photon regime, the structure of the problem is organized at two levels. On the one hand, the Keyl–Werner result identifies the representations whose normalized shape  $\bar{\Lambda}$  concentrates around the spectrum  $\boldsymbol{\lambda} = \text{spec } \rho_{\bar{n},\Gamma}$  [6]. On

the other hand, within a fixed irrep, the matrix

$$\rho_\Lambda(V) = V[\bar{\Lambda}]V^\dagger \quad (5.1)$$

determines through its diagonal the macroscopic weight  $Q(V)\bar{\Lambda}$  and, after the action of  $Ut_\phi$ , produces the sectorial classical map of Eq. (4.13). In the dominant sectors, this construction is directly related to the effective single-particle matrix and to the map obtained for partially distinguishable interference [3].

Multiplicity introduces an additional level. For  $SU(3)$ , a projected frame state has the expansion

$$|W, V\rangle = \sum_j c_j(V) |W, j\rangle, \quad (5.2)$$

so that a fixed occupation does not determine the distribution over the isospin values. The results of Chapter 3 further show that frames with the same matrix  $Q(V)$  can select different directions within the same weight space. This suggests studying what information in  $\rho_\Lambda(V)$ , beyond its diagonal, controls the concentration over the labels that resolve the multiplicity. The off-diagonal elements depend on information from the frame that does not appear in  $Q(V)$ . In  $SU(3)$ , the operators that resolve isospin are constructed from diagonal and ladder generators, which provides a concrete route for investigating whether this information also controls the concentration in  $j$ . Establishing this relation in the limit  $N \rightarrow \infty$  constitutes one of the central problems of the doctoral research.

The doctoral research will start from the already established relation between  $\rho_\Lambda(V)$  and concentration in weights to investigate what additional information in this matrix makes it possible to characterize the asymptotic concentration over the labels that resolve the multiplicity. Based on this characterization, we will study how this structure is incorporated into the density  $\sigma_{\Lambda, \mathbf{n}}(\Gamma)$  in the dominant irreducible sectors and how it is reflected in multiphotonic interference probabilities with partial indistinguishability.

## 5.2 Objectives

### 5.2.1 General objective

Determine to what extent the matrix  $\rho_\Lambda(V)$  makes it possible to characterize the asymptotic concentration within the multiplicity spaces and establish what additional information, beyond that associated with the weights, remains relevant in this limit. Based on this characterization, analyze how this structure is related to the dynamics within the irreducible sectors and to the effect of partial indistinguishability on multiphotonic interference probabilities.

### 5.2.2 Specific objectives

1. Characterize in  $SU(3)$  the joint distribution of weight and isospin associated with the orbit states and, after projecting onto a fixed weight, determine the center and dispersion of  $|c_j(V)|^2$  in the regime  $N \rightarrow \infty$ .
2. Relate these quantities to the elements and invariants of  $\rho_\Lambda(V)$  and determine how the choice of frame, while keeping the spectrum–diagonal condition fixed, modifies the concentration within the multiplicity space.
3. Extend this analysis to  $U(M)$  through the Gelfand–Tsetlin chain, identifying the labels that resolve the multiplicity spaces and studying the conditions under which their distributions exhibit asymptotic concentration.
4. Incorporate this description into the density  $\sigma_{\Lambda, \mathbf{n}}(\Gamma)$  and the sum over sectors in Eq. (1.10), in order to obtain an asymptotic approximation of the physical probabilities that retains the information associated with multiplicity.

### 5.3 Methodology

The first step will be developed in  $SU(3)$ , where multiplicity can be resolved explicitly in the basis  $|W, j\rangle$ . We will start from the exact coefficients  $c_j(V)$  obtained in Sec. 3.2 to construct the joint distribution of weight and isospin before projection and the conditional distribution over  $j$  after fixing  $W$ . Its moments and large- $N$  behavior will be compared with quantities constructed from  $\rho_\Lambda(V)$ . The spectral invariants, such as  $\text{Tr } \rho_\Lambda^k$  and  $\det \rho_\Lambda$ , will impose constraints among the variables associated with different  $SU(2)$  subgroups and will make it possible to study which frames produce more concentrated distributions within a weight space.

The second stage will seek to distinguish which aspects of the result depend on particular properties of  $SU(3)$  from those that persist for  $U(M)$ . The Gelfand–Tsetlin chain will provide the labels that resolve the successive multiplicities. We will study generating functions, moments, and stationary-point approximations to establish whether these labels exhibit a concentration controlled by the matrix  $\rho_\Lambda(V)$  and its spectral constraints. The symbolic and numerical calculations will be used to verify the analytical expressions at finite photon number and to identify the regimes in which the expected concentration appears.

Finally, we will return to the physical density  $\sigma_{\Lambda, \mathbf{n}}(\Gamma)$  defined in Eq. (1.9) and discussed in Sec. 2.3. Rather than identifying it with a single frame state, we will analyze its elements in a basis that resolves the multiplicity and their contribution to the trace in Eq. (1.11). The Keyl–Werner concentration will make it possible to restrict the sum to the dominant sectors, whereas the sectorial map of Eq. (4.13) will determine the macroscopic zero-rate region. The objective will be to determine how the distribution within the multiplicity spaces is reflected in the prefactors and in the fine structure of the probability, and whether it introduces additional corrections to the macroscopic geometry obtained for the frame states. As an extension of the regular analysis, we will study regimes in which the Hessian becomes degenerate and caustics appear, as well as the possible contribution of complex stationary points.

### 5.4 Ethical considerations

The project is theoretical and computational in nature and does not involve human participants, animals, or the collection of personal information. Numerical calculations will be used to verify analytical results and will be documented so that the configurations, parameters, and procedures used can be reproduced. The presentation of the results will explicitly distinguish between analytical proofs, numerical evidence, and hypotheses that remain open.

### 5.5 Tentative schedule

| Period      | Main activity                                                                                                                                                       |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Semester 6  | Joint concentration of weight and isospin in $SU(3)$ ; calculation of moments, dispersion, and comparison with the exact results for the frame states.              |
| Semester 7  | Characterization of the dependence on the frame and of the constraints imposed by the invariants of $\rho_\Lambda(V)$ ; beginning of the extension to $U(M)$ .      |
| Semester 8  | Generalization through the Gelfand–Tsetlin chain and analytical and numerical validation for higher-dimensional representations and weights.                        |
| Semester 9  | Incorporation of multiplicity into $\sigma_{\Lambda, \mathbf{n}}(\Gamma)$ and the sectorial probabilities; analysis of prefactors in the regular stationary regime. |
| Semester 10 | Extensions of the asymptotic analysis, integration of the results, and writing and defense of the doctoral thesis.                                                  |



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